



Stock Trading Platform

CodeAlpha Java Programming Internship

Project Overview

The **Stock Trading Platform** is a console-based Java application that simulates basic stock market operations. It allows users to view market data, buy stocks, sell stocks, and manage their investment portfolio through an interactive menu-driven interface.

This project was developed as part of the **CodeAlpha Java Programming Internship** to demonstrate the use of Object-Oriented Programming concepts and efficient data structures.

Features

- View available stocks and their prices
 - Dynamic stock price updates
 - Buy stocks
 - Sell stocks
 - Portfolio management
 - Balance tracking
 - Interactive console interface
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Technologies Used

- Java
 - Object-Oriented Programming (OOP)
 - HashMap
 - Random Class
 - Scanner Class
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OOP Concepts Implemented

- Classes and Objects
 - Encapsulation
 - Composition
 - Method Overriding
-

Project Structure

```
StockTradingPlatform/  
|
```

```
|— Main.java
|— Stock.java
|— Portfolio.java
└— TradingPlatform.java
```

How to Run

Compile the project

```
javac *.java
```

Run the application

```
java Main
```

Sample Menu

```
1. View Market
2. Buy Stock
3. Sell Stock
4. Portfolio
5. Exit
```

Time Complexity

Operation	Complexity
Buy Stock	$O(1)$
Sell Stock	$O(1)$
Search Stock	$O(1)$
Display Market	$O(n)$
Display Portfolio	$O(k)$

Where:

- **n** = Number of available stocks
 - **k** = Number of owned stocks
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Future Enhancements

- Transaction History
 - File Storage
 - User Authentication
 - Graphical User Interface (GUI)
 - Database Integration
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Internship

Organization: CodeAlpha

Domain: Java Programming

Project: Stock Trading Platform

Author

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